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Brief Paper

Interval consensus over random networks☆

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ABSTRACT

This paper considers the interval consensus problems of discrete-time multi-agent systems over random interaction networks, where each agent can impose a lower and an upper bound, i.e., a local constraint interval, on the achievable consensus value. We show that if the intersection of the intervals is nonempty, it holds as a sure event that the states of all the agents converge to a common value inside that intersection, i.e., the interval consensus can be achieved almost surely. Convergence analysis is performed through developing a robust consensus analysis of random networks in view of a martingale convergence lemma. Numerical examples are also exhibited to verify the validity of the theoretical results.

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1. Introduction

Consensus algorithms have become a fundamental building block in the design of distributed control and optimization protocols (Jadbabaie, Lin, & Morse, 2003; Mesbahi & Egerstedt, 2010; Moreau, 2005; Olfati-Saber & Murray, 2004; Qin, Gao, & Yu, 2014; Qin, Ma, Zheng, Gao, & Kang, 2017; Tsitsiklis, 1984), where applications have been found in formation of robotics (Qin, Wang, Kang and Liu, 2018; Ren & Sorensen, 2008; Wang, Huang, Wen, & Fan, 2014), gradient algorithm for cooperative convex optimization (Nedić & Ozdaglar, 2009; Nedić, Ozdaglar, & Parrilo, 2010), and economic dispatch solutions in power networks and smart grid (Qin, Wan, Yu, Li and Li, 2018; Yang et al., 2017). In a standard consensus seeking problem, nodes update their states as a convex combination of their neighbors' states, where under rather general connectivity conditions the network states asymptotically agree on a common value related to the initial states and possibly also to the network structure (Mesbahi & Egerstedt, 2010). Such algorithms turn out to be resilient to node communication failures and noises (Kar & Moura, 2009; Shi & Johansson, 2013).

However, in all of these problems, only a part of the nodes have the authorities to determine certain values of consensus, or to impose that the consensus is restricted within an admissible region. This is a drastic limitation in certain contexts. For instance, in the economic dispatch problem of smart grid (Qin, Wan et al., 2018; Yang et al., 2017), whose aim is to minimize the total generation cost while satisfying the demand constraint, the powers generated through making an agreement on the incremental cost cannot exceed the capability of each generator. In an opinion dynamics context, each node might agree on a common opinion only if this is acceptable by itself. In a robotic formation, a robot might be able to remain aligned with the formation only if the consensus velocity is achievable by all the robots. These examples motivate investigation of consensus problems with node state constraints.

Consensus problems with constraints have been studied from different perspectives in the literature. Lee and Nedić (2013) and Nedić et al. (2010) deal with the distributed constrained convex optimization problems for multiple agents connected over a network and propose a projected consensus algorithm, where each agent projects the combination of the values received from the neighbors on its own constraint set. The discarded consensus algorithm in Liu and Chen (2012), where each agent discards the states of neighbors if these states are outside its constraint, solves the consensus problems of multi-agent systems with state constraint. In the opinion dynamics literature (Blondel, Hendrickx, & Tsitsiklis, 2009), bounded confidence models are proposed, where the states are ignored in the state update of an agent if they are more distant than a certain threshold. These models

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can usually produce clusters of opinions, and a local consensus value within each cluster. Recently, an interval consensus problem is considered in Fontan, Shi, Hu, and Altafini (2017, 2019), where each agent is given the possibility of limiting the interval of values in which a consensus value can be accepted. More specifically, Fontan et al. (2017) considers the interval consensus problems for continuous-time and discrete-time first-order multi-agent systems over directed and strongly connected interaction graph. It is shown that if such state constraints are implemented by saturating the value transmitted to the neighboring nodes, the resulting consensus problem must converge to the intersection of the intervals imposed by the individual agents. The work in Fontan et al. (2019), established on Fontan et al. (2017), also discusses the case that the intersection of the intervals is an empty set. It is proved that in that case there exists at least one equilibrium, and in fact the possible equilibria are locally stable if the constraints are satisfied or dissatisfied at the same time among all nodes. For graphs with sufficient sparsity it is further proved that there is a unique equilibrium that is globally attractive if the constraint intervals are pairwise disjoint.

Random node interactions naturally occur due to the nature of communication links, human decision, or inherent randomization of protocols (Boyd, Ghosh, Prabhakar, & Shah, 2006; Dimakis, Kar, Moura, Rabbat, & Scaglione, 2010; Shi & Anderson, 2015; Tahbaz-Salehi & Jadbabaie, 2008, 2010; Touri & Nedić, 2011). With mean-square analysis, convergence can be shown for consensus over i.i.d. random networks (Boyd et al., 2006). More involved analysis establishes the zero–one property of general consensus computations with random node updates (Shi & Anderson, 2015; Tahbaz-Salehi & Jadbabaie, 2008; Touri & Nedić, 2011). It is of interest to understand the effect of random node interactions in interval consensus algorithms. On one hand, the motivating examples from social interactions suggest that random graphs suit more to the background. On the other hand, it is a non-trivial problem from the theoretical point of view due to the nonlinearity of the node updates induced by the constraint intervals at individual node. In this paper, we prove that interval consensus can be achieved almost surely under a random network model where each agent restricts the values transmitted to its neighbors in a fixed interval. To facilitate the analysis, we also develop theories for robust consensus under random setup, which might be of independent use. Numerical results are provided to demonstrate our theoretical findings.

The remainder of this paper is organized as follows. Section 2 presents the interval consensus model and the main result. Robust consensus problems over random networks are considered in Section 3 in order to derive the proof of the main result in Section 4. The obtained results are verified through simulation examples in Section 5 and then summarized in Section 6.

2. Model and results

2.1. The interval consensus model

Consider a network of n nodes indexed in the set $V = \{1, \dots, n\}$. The underlying interaction structure of the network is described by an undirected graph $G = (V, E)$, which is assumed to be connected. Here $E \subseteq V \times V$ is an unordered edge set and $(i, j) \in E$ means that there are communication links between node i and node j. Denoted by $N_i = \{j \in V : (i, j) \in E\}$ the set of neighbors of node i in the underlying graph G. The set of all subgraphs of G is denoted as $S = \{G_k = (V, E_k)\}_{k\in\{1,\dots,c\}}$ with $c > 0$ as a finite integer and $\cup_{k=1}^{c} E_k = E$. Let μ be a finite probability measure over S satisfying $\mu(G_k) \geq \alpha, k \in \{1, \dots, c\}$ for some $\alpha > 0$.

The communication among the nodes at time $t = 0, 1, 2, \dots$ is drawn from S by the probability measure μ independently with other time and node states, and the resulting random graph process is $g_t = G_{\sigma(t)} \in S, t = 0, 1, 2, \dots$ with $\sigma : \mathbb{N} \to \{1, \dots, c\}$ being the random variable that determines the network topology. The neighbor set of node i at time t is denoted as $N_i(t) = \{j : (i, j) \in E_{\sigma(t)}\}$. In addition, we let $\mathcal{F}_t$ be the σ-algebra generated by $g_0, \dots, g_t$ for $t \geq 0$.

Each node i holds a state $x_i(t)$ at time $t \in \mathbb{N}$. Furthermore, node i also holds an interval $I_i = [p_i, q_i]$ determining the range of the values node i can transmit. More specifically, the node sends $\psi_i(x_i(t))$ to its neighbors instead of $x_i(t)$, where

$$\psi_i(z) = \begin{cases} p_i, & \text{if} \quad z < p_i \\ z, & \text{if} \quad p_i \leq z \leq q_i \\ q_i, & \text{if} \quad z > q_i \end{cases}. \tag{1}$$

With the above notations, the interval consensus algorithm of node i is described by

$$\begin{aligned} x_i(t+1) &= x_i(t) + \varepsilon \sum_{j\in N_i(t)} a_{ij}(\psi_j(x_j(t)) - x_i(t)) \\ &= (1 - \varepsilon \sum_{j\in N_i(t)} a_{ij}) x_i(t) + \varepsilon \sum_{j\in N_i(t)} a_{ij}\psi_j(x_j(t)), \end{aligned} \tag{2}$$

where $a_{ij} = a_{ji} > 0$ is the weight of edge (i, j), and $\varepsilon > 0$ is a step size.

Remark 1. Note that in algorithm (2), each node received projected states from its neighbors, which is different with the algorithms considered in Blondel et al. (2009), Lee and Nedić (2013), Liu and Chen (2012) and Nedić et al. (2010) where the actual neighbors' states are utilized. This is because algorithm (2) is mainly motivated by the phenomenon in opinion dynamics, for example, achieving a price agreement among shareholders (Fontan et al., 2017, 2019), that individuals would rather express the opinions accepted by them.

2.2. Main results

Denote by $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))^T$ the network state. The following theorem suggests that if the intersection of the intervals imposed by the nodes is nonempty, the states of the nodes will converge to a common value inside that intersection almost surely.

Theorem 1. *Suppose $\cap_{m=1}^{n} I_m \neq \varnothing$ and $2\varepsilon \sum_{j\in N_i} a_{ij} < 1$ for all $i \in V$. Then for any initial value $\mathbf{x}(0)$, there is a random variable $y_*(\mathbf{x}(0)) \in \cap_{m=1}^{n} I_m$ such that*

$$\mathbb{P}(\lim_{t\to\infty} x_i(t) = y_*, i \in V) = 1.$$

Note that the condition $\cap_{m=1}^{n} I_m \neq \varnothing$ is equivalent to $\cap_{m=1}^{n} I_m = [p_*, q_*]$ with $p_* = \max_{i\in V} p_i$, $q_* = \min_{i\in V} q_i$.

Theorem 1 is consistent with the standard consensus algorithm in terms of the almost sure convergence statement (Boyd et al., 2006; Dimakis et al., 2010; Shi & Anderson, 2015; Touri & Nedić, 2011). However, the nonlinearity in the interval constraint function $\psi_i(\cdot)$ fundamentally changes the nature of the algorithm, where standard mean-square analysis no longer applies (Boyd et al., 2006; Dimakis et al., 2010). This forces us to develop a robust consensus theory for random graphs as a preliminary step, which is analogous to the work of Shi and Johansson (2013) for deterministic networks and might be of independent interest. We have employed carefully structured random sequences which happen to be martingales for such development.

When $\cap_{m=1}^{n} I_m = \varnothing$, the states of all the nodes will in general fail to asymptotically converge, even to different values with each other. This is in contrast with the deterministic setting in Fontan et al. (2019), where equilibria are often locally stable. Note that the local stability is easily destroyed over random networks due to sampling of the interaction graphs.

3. Robust consensus of random networks

In order to establish the presented theorem, we first generalize the notion of robust consensus for deterministic networks (Shi & Johansson, 2013) to the random case.

Proposition 1. *Consider the network dynamics below.*

$$x_i(t+1) = x_i(t) + \varepsilon \sum_{j\in N_i(t)} a_{ij}(x_j(t) - x_i(t)) + w_i(t),$$
$$i \in V, t \in \mathbb{N}, \tag{3}$$

where $2\varepsilon \sum_{j\in N_i} a_{ij} < 1$. *If* $w_i(t)$ *is* $\mathcal{F}_t$ *measurable with* $|w_i(t)| \le w^*$ *for* $w^* > 0$ *and* $\mathbb{P}(\lim_{t\to\infty} w_i(t) = 0) = 1$, *then for any initial states* $\mathbf{x}(0)$, *there holds that*

$$\mathbb{P}(\lim_{t\to\infty} \max_{i,j\in V} |x_i(t) - x_j(t)| = 0) = 1.$$

In order to prove Proposition 1, the following two lemmas are necessary. Lemma 1 is the well known Brauer's Theorem in Horn and Johnson (2012) and Lemma 2 in Polyak (1987), Robbins and Siegmund (1971) is a martingale convergence lemma describing the convergence of a certain sequence of random variables.

Lemma 1. *Let* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$, *and* $A \in \mathbb{R}^{n\times n}$. *Suppose that* $A\mathbf{x} = \lambda\mathbf{x}$ *and let the eigenvalues of* A *be* $\lambda, \lambda_2, \ldots, \lambda_n$, *then the eigenvalues of* $A + \mathbf{x}\mathbf{y}^T$ *are* $\lambda + \mathbf{y}^T\mathbf{x}, \lambda_2, \ldots, \lambda_n$.

Lemma 2. *Let* $\{v_t\}_{t\in\mathbb{N}}$ *be a sequence of non-negative random variables with* $\mathbb{E}\{v_0\} < \infty$. *Assume that for any* $t \in \mathbb{N}$, $\mathbb{E}\{v_{t+1}|v_0, \ldots, v_t\} \le (1-\xi_t)v_t + \theta_t$, *where* $\{\xi_t\}_{t\in\mathbb{N}}$ *and* $\{\theta_t\}_{t\in\mathbb{N}}$ *are two sequences of non-negative numbers satisfying* $\forall t \in \mathbb{N}$, $0 \le \xi_t \le 1$, $\sum_{t=0}^{\infty} \xi_t = \infty$, $\sum_{t=0}^{\infty} \theta_t < \infty$, *and* $\lim_{t\to\infty} \theta_t/\xi_t = 0$. *Then,* $\lim_{t\to\infty} v_t = 0$ *almost surely.*

3.1. Proof of Proposition 1

One can rewrite (3) in a compact form as

$$\mathbf{x}(t+1) = (I - \varepsilon L_t)\mathbf{x}(t) + \mathbf{w}(t),$$

where L_t is the Laplacian of the random graph g_t and $\mathbf{w}(t) = (w_1(t), \ldots, w_n(t))^T$. Denote $J = \frac{1}{n}\mathbf{1}\mathbf{1}^T$, where $\mathbf{1}$ is the n-dimensional all-one vector. Introduce $\mathbf{y}(t) = (I-J)\mathbf{x}(t)$, one has

$$\begin{aligned}\mathbf{y}(t+1) &= (I-J)\mathbf{x}(t+1)\\ &= (I - \varepsilon L_t - J)\mathbf{x}(t) + (I-J)\mathbf{w}(t)\\ &= (I - \varepsilon L_t - J)(I-J)\mathbf{x}(t) + (I-J)\mathbf{w}(t)\\ &= M_t\mathbf{y}(t) + (I-J)\mathbf{w}(t),\end{aligned}$$

where $M_t = I - \varepsilon L_t - J$, since $L_tJ = JL_t = 0$.

We then establish a useful lemma regarding the random matrix M_t in the following.

Lemma 3. *Suppose* $2\varepsilon \sum_{j\in N_i} a_{ij} < 1$ *for all* $i \in V$. *Then*

$$0 < \lambda_{\max}(\mathbb{E}\{M_t^2\}) < 1.$$

Proof. From the definition of the graph g_t, there holds for every realization of g_t that

$$1 - 2\varepsilon \sum_{j\in N_i(t)} a_{ij} > 0$$

given $2\varepsilon \sum_{j\in N_i} a_{ij} < 1$. Since Gershgorin's Disc Theorem implies that all eigenvalues of $I - \varepsilon L_t$ are located in the union of the n discs given by

$$\bigcup_{i=1}^{n}\Big\{z \in \mathbb{C} : |z - (1 - \varepsilon \sum_{j\in N_i(t)} a_{ij})| \le \varepsilon \sum_{j\in N_i(t)} a_{ij}\Big\},$$

one can conclude that $0 < \lambda(I - \varepsilon L_t) \le 1$. Furthermore, $\mathbf{1}$ is the right eigenvector of $I - \varepsilon L_t$ associated with the eigenvalue 1, which together with Lemma 1 yields that $0 < \lambda(M_t) \le 1$. One can therefore obtain that $M_t^2 \le M_t$ holds as a sure event. Thus, $\mathbb{E}\{M_t^2\} \le \mathbb{E}\{M_t\}$.

In addition,

$$\mathbb{E}\{M_t\} = I - \varepsilon\mathbb{E}\{L_t\} - J = I - \varepsilon L^*(G) - J,$$

where $L^*(G)$ is a weighted Laplacian of the graph G. Note that since G is connected, one obtains that $I - \varepsilon L^*(G)$ has a unique eigenvalue 1 corresponding to eigenvector $\mathbf{1}$. This allows us to conclude $0 < \lambda(I - \varepsilon L^*(G) - J) < 1$ according to Lemma 1. Therefore, we have proved $\lambda_{\max}(\mathbb{E}\{M_t^2\}) \le \lambda_{\max}(\mathbb{E}\{M_t\}) < 1$. The part for $\lambda_{\max}(\mathbb{E}\{M_t^2\}) > 0$ is obvious. This proves the lemma. ■

We can now proceed to prove the proposition. For brevity, denote $1 - \lambda^* = \lambda_{\max}(\mathbb{E}\{M_t^2\})$, where $0 < \lambda^* < 1$.

By applying Cauchy–Schwarz inequality, one has

$$\begin{aligned}&\mathbb{E}\{|\mathbf{y}(t+1)|^2\big||\mathbf{y}(t)|^2, \ldots, |\mathbf{y}(0)|^2\}\\ &\quad= \mathbf{y}^T(t)\mathbb{E}\{M_t\}^2\mathbf{y}(t) + 2\mathbf{y}^T(t)\mathbb{E}\{M_t\}(I-J)\mathbf{w}(t)\\ &\qquad+ \mathbf{w}^T(t)(I-J)\mathbf{w}(t)\\ &\quad\le (1-\lambda^*)|\mathbf{y}(t)|^2 + 2|\mathbf{y}(t)|\cdot|\mathbf{w}(t)| + |\mathbf{w}(t)|^2\\ &\quad\le [1 - (\lambda^* - |\mathbf{w}(t)|)]|\mathbf{y}(t)|^2 + |\mathbf{w}(t)| + |\mathbf{w}(t)|^2.\end{aligned}$$

Now since $\mathbf{w}(t)$ is $\mathcal{F}_t$-measurable, there are two deterministic functions h_1 and h_2 such that $|\mathbf{w}(t)| = h_1(\mathbf{y}(0), \ldots, \mathbf{y}(t))$ and $|\mathbf{w}(t)| + |\mathbf{w}(t)|^2 = h_2(\mathbf{y}(0), \ldots, \mathbf{y}(t))$. Then one can obtain that

$$\begin{aligned}&\mathbb{E}\{|\mathbf{y}(t+1)|^2\big||\mathbf{y}(t)|^2, \ldots, |\mathbf{y}(0)|^2\}\\ &\quad\le [1 - (\lambda^* - h_1(t))]|\mathbf{y}(t)|^2 + h_2(t),\end{aligned}$$

where $P(\lim_{t\to\infty} h_1(t) = 0) = 1$ and $P(\lim_{t\to\infty} h_2(t) = 0) = 1$.

As a result, there exists $T > 0$ such that $h_1(t) \le \lambda^*$, $h_2(t) \le \lambda^*$, $\forall t \ge T$. We can then obtain that $0 \le \lambda^* - h_1(t) \le 1$, $\forall t \ge T$, $\sum_{t=T}^{\infty}(\lambda^* - h_1(t)) = \infty$, $\sum_{t=T}^{\infty} h_2(t) < \infty$, and $\lim_{t\to\infty} h_2(t)/(\lambda^* - h_1(t)) = 0$. Furthermore,

$$\begin{aligned}&\mathbb{E}\{|\mathbf{y}(t+1)|^2\big||\mathbf{y}(t)|^2, \ldots, |\mathbf{y}(0)|^2\}\\ &\quad= \mathbb{E}\{|\mathbf{y}(t+1)|^2\big||\mathbf{y}(t)|^2, \ldots, |\mathbf{y}(T)|^2\}\\ &\quad\le [1 - (\lambda^* - h_1(t))]|\mathbf{y}(t)|^2 + h_2(t), t \ge T.\end{aligned}$$

We can thus invoke Lemma 2 to conclude that $\mathbb{P}(\lim_{t\to\infty} \mathbf{y}(t) = 0) = 1$, i.e., $\mathbb{P}(\lim_{t\to\infty} \max_{i,j\in V} |x_i(t) - x_j(t)| = 0) = 1$. ■

4. Proof of main result

In this section, we will follow the main argument of Fontan et al. (2017, 2019), but adopt the analysis to the considered probabilistic setting to prove Theorem 1.

Introduce $H(\mathbf{x}(t)) = \max\{\max_{i\in V} x_i(t), q_*\}$ and $h(\mathbf{x}(t)) = \min\{\min_{i\in V} x_i(t), p_*\}$. Then, we have the following lemma related to $H(\mathbf{x}(t))$ and $h(\mathbf{x}(t))$.

Lemma 4. $\{H(\mathbf{x}(t+1)) \le H(\mathbf{x}(t))\}$ *and* $\{h(\mathbf{x}(t+1)) \ge h(\mathbf{x}(t))\}$ *are sure events if* $\varepsilon \sum_{j\in N_i} a_{ij} < 1$.

Proof. Define $I_a^+(\cdot)$ with $I_a^+(z) = z$ if $z > a$ and $I_a^+(z) = a$ otherwise. Then, one can obtain that

$$\begin{aligned}H(\mathbf{x}(t)) &= I_{q_*}^+(\max_{i\in V} x_i(t+1))\\ &= I_{q_*}^+\Big(\max_{i\in V}\big((1 - \varepsilon \sum_{j\in N_i(t)} a_{ij})x_i(t)\\ &\quad+ \varepsilon \sum_{j\in N_i(t)} a_{ij}\psi_j(x_j(t))\big)\Big)\end{aligned}$$

$$\le I_{q_*}^+\Big(\max_{i\in V}\big((1-\varepsilon\sum_{j\in N_i(t)}a_{ij})I_{q_*}^+(\max_{j\in V}x_j(t))$$
$$+\varepsilon\sum_{j\in N_i(t)}a_{ij}I_{q_*}^+(\max_{j\in V}x_j(t))\big)\Big)$$
$$=H(\mathbf{x}(t)), \tag{4}$$

regardless of the choice of g_t (therefore of $N_i(t)$ as well). This proves that $\{H(\mathbf{x}(t+1)) \le H(\mathbf{x}(t))\}$ is a sure event. The other part follows a similar argument. ■

Based on Lemma 4, we establish the lemma below.

Lemma 5. *Let $\cap_{m=1}^n I_m \neq \varnothing$ and $\varepsilon\sum_{j\in N_i}a_{ij} < 1$, there hold $\mathbb{P}(\lim_{t\to\infty}H(\mathbf{x}(t))=q_*)=1$ and $\mathbb{P}(\lim_{t\to\infty}h(\mathbf{x}(t))=p_*)=1$.*

Proof. According to Lemma 4, it is apparent that there are two random variable H_* and h_* such that

$$\mathbb{P}(\lim_{t\to\infty}H(\mathbf{x}(t))=H_*,\ \lim_{t\to\infty}h(\mathbf{x}(t))=h_*)=1.$$

Obviously from the definition of $H(\mathbf{x}(t))$ and $h(\mathbf{x}(t))$, one can also know that $H_*\ge q_*\ge p_*\ge h_*$ holds almost surely. Then we need to prove that $\mathbb{P}(H_*=q_*)=1$ and $\mathbb{P}(h_*=p_*)=1$.

Suppose for the sake of a contradiction argument that $\mathbb{P}(p_*>h_*)>0$. Then we can fix a time s and take a node i_0 with $x_{i_0}(s)=\min_{j\in V}x_j(s)\le h_*<p_*$ conditioned on the event $\{p_*>h_*\}$. Such a node may exist since $\mathbb{P}(p_*>h_*=\lim_{t\to\infty}\min\{\min_{j\in V}x_j(t),p_*\})>0$. Denote $l_0=\{j: x_j(s)=x_{i_0}(s)\}$ and define $l_{k+1}=l_k\cup\check{l}_{k+1}$ iteratively, where $\check{l}_{k+1}=\{j\notin l_k: j\in N_i(s+k), i\in l_k\}$ represents the set of the nodes outside l_k, which communicate with the nodes in l_k at time $s+k$. In the following, we will prove by induction that

$$x_i(s+k)\le\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s)),\ \forall i\in l_k,\ k\in\mathbb{N}^+,$$

where $\theta=\min\{\varepsilon\min_{(i,j)\in E}\{a_{ij}:a_{ij}\neq 0\},\min_i\{1-\varepsilon\sum_{j\in N_i}a_{ij}\}\}$.

Note that the nodes in l_k can be classified into three clusters, i.e., $l_k=\bar{l}_k\cup\hat{l}_k\cup\check{l}_k$, where $\bar{l}_k=\{i:|N_i(s+k-1)|=\varnothing\}$ and $\hat{l}_k=\{i:|N_i(s+k-1)|\neq\varnothing\}$ represent the sets of nodes in l_{k-1} without and with neighbors at time $s+k-1$, and $\check{l}_k=\{j\notin l_{k-1}: j\in N_i(s+k-1), i\in l_{k-1}\}$ is defined previously. For $k=1$, the following three cases are investigated, respectively:

(a) For $i\in\bar{l}_1\subseteq l_0$, it is apparent that

$$x_i(s+1)=x_i(s)\le p_*\le\theta p_*+(1-\theta)I_{q_*}^+(\max_{j\in V}x_j(s)) \tag{5}$$

since $I_{q_*}^+(\max_{j\in V}x_j(s))\ge q_*\ge p_*$.

(b) For $i\in\hat{l}_1\subseteq l_0$, one has

$$x_i(s+1)=(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})x_i(s)+\varepsilon\sum_{j\in N_i(s)}a_{ij}\psi_j(x_j(s))$$
$$\le(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})p_*+(\varepsilon\sum_{j\in N_i(s)}a_{ij})I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$\le\theta p_*+(1-\theta)I_{q_*}^+(\max_{j\in V}x_j(s)), \tag{6}$$

since $I_{q_*}^+(\max_{j\in V}x_j(s))\ge p_*$ and $(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})\ge\theta$.

(c) For $i\in\check{l}_1$, assume that $i'\in l_0$ is the neighbor of i, one has

$$x_i(s+1)=(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})x_i(s)+\varepsilon\sum_{j\in N_i(s)}a_{ij}\psi_j(x_j(s))$$
$$=(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})x_i(s)+\varepsilon\sum_{j\neq i'\in N_i(s)}a_{ij}\psi_j(x_j(s))$$
$$+\varepsilon a_{ii'}\psi_{i'}(x_{i'}(s))$$
$$\le(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$+\varepsilon\sum_{j\neq i'\in N_i(s)}a_{ij}I_{q_*}^+(\max_{j\in V}x_j(s))+\varepsilon a_{ii'}p^*$$
$$\le\theta p_*+(1-\theta)I_{q_*}^+(\max_{j\in V}x_j(s)). \tag{7}$$

In conclusion, one can obtain that $x_i(s+1)\le\theta^1 p_*+(1-\theta^1)I_{q_*}^+(\max_{j\in V}x_j(s))$, $\forall i\in l_1$.

Next, we suppose $x_i(s+k)\le\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))$, then

(a) For $i\in\bar{l}_{k+1}\subseteq l_k$, it is apparent that

$$x_i(s+k+1)=x_i(s+k)\le\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$\le\theta^{k+1}p_*+(1-\theta^{k+1})I_{q_*}^+(\max_{j\in V}x_j(s)). \tag{8}$$

(b) For $i\in\hat{l}_{k+1}\subseteq l_k$, one has

$$x_i(s+k+1)=(1-\varepsilon\sum_{j\in N_i(s+k)}a_{ij})x_i(s+k)$$
$$+\varepsilon\sum_{j\in N_i(s+k)}a_{ij}\psi_j(x_j(s+k))$$
$$\le(1-\varepsilon\sum_{j\in N_i(s)}a_{ij})\big[\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))\big]$$
$$+(\varepsilon\sum_{j\in N_i(s)}a_{ij})I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$=\theta^k p_*+\theta^k(\varepsilon\sum_{j\in N_i(s)}a_{ij})(I_{q_*}^+(\max_{j\in V}x_j(s))-p_*)$$
$$+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$\le\theta^k p_*+\theta^k(1-\theta)(I_{q_*}^+(\max_{j\in V}x_j(s))-p_*)$$
$$+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$=\theta^{k+1}p_*+(1-\theta^{k+1})I_{q_*}^+(\max_{j\in V}x_j(s)). \tag{9}$$

(c) For $i\in\check{l}_{k+1}$, assume that $i'\in l_k$ is the neighbor of i, one has

$$x_i(s+k+1)$$
$$=(1-\varepsilon\sum_{j\in N_i(s+k)}a_{ij})x_i(s+k)$$
$$+\varepsilon\sum_{j\neq i'\in N_i(s+k)}a_{ij}\psi_j(x_j(s+k))+\varepsilon a_{ii'}\psi_{i'}(x_{i'}(s+k))$$
$$\le(1-\varepsilon a_{ii'})I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$+\varepsilon a_{ii'}\big[\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))\big]$$
$$\le(1-\theta)I_{q_*}^+(\max_{j\in V}x_j(s))$$
$$+\theta\big[\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))\big]$$
$$=\theta^{k+1}p_*+(1-\theta^{k+1})I_{q_*}^+(\max_{j\in V}x_j(s)). \tag{10}$$

Therefore, from the assumption that $x_i(s+k)\le\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s))$, one can obtain $x_i(s+k+1)\le\theta^{k+1}p_*+(1-\theta^{k+1})I_{q_*}^+(\max_{j\in V}x_j(s))$, $\forall i\in l_{k+1}$, which can further by induction result in

$$x_i(s+k)\le\theta^k p_*+(1-\theta^k)I_{q_*}^+(\max_{j\in V}x_j(s)),\ \forall i\in l_k,\ k\in\mathbb{N}^+.$$

Note that since the underlying interaction graph G is connected, there exists a time instant $s+\tau > s$ such that $l_k = V, \forall k \geq \tau$ holds as a sure event, leading to that

$$x_i(t) \leq \theta^{t-s} p_* + (1-\theta^{t-s}) I^+_{q_*}(\max_{j\in V} x_j(s)), \forall i \in V, t \geq s+\tau.$$

At this point, the following two cases are considered, respectively.

(a) Suppose $p_* < H_*$. Then for sufficiently large s, $I^+_{q_*}(\max_{j\in V} x_j(s))$ will be so close to H_* that $\theta^\tau p_* + (1-\theta^\tau) I^+_{q_*}(\max_{j\in V} x_j(s)) < H_*$. Therefore, one can conclude that $x_i(t) < H_*, \forall i \in V, \forall t \geq s+\tau$ for all s that are sufficiently large. From the definition of $H(\mathbf{x}(t))$ and H_*, it can be only obtained that $H_* = q_*$. As a result, $x_i(t) < q_*, \forall i \in V, \forall t \geq s+\tau$ for sufficiently large s, which implies that there exists $T > 0$ such that $x_i(t) \leq I^+_{p_*}(\max_{j\in V} x_j(t))$, $\psi_i(x_i(t)) \leq I^+_{p_*}(\max_{j\in V} x_j(t))$ for all j and all $t \geq T$.

In this way, the term $I^+_{q_*}$ in (4) can be replaced by $I^+_{p_*}$ for $s > T$. Therefore, one can obtain that $I^+_{p_*}(\max_{j\in V} x_j(s))$ is non-increasing for $s > T$, which means that $I^+_{p_*}(\max_{j\in V} x_j(s+k)) \leq I^+_{p_*}(\max_{j\in V} x_j(s))$ for any $k > 0$.

Therefore, one can replace the term $I^+_{q_*}(\max_{j\in V} x_j(s))$ by $I^+_{p_*}(\max_{j\in V} x_j(s))$ in (5)–(10), which can further yield that $x_i(t) \leq \theta^{t-s} p_* + (1-\theta^{t-s}) I^+_{p_*}(\max_{j\in V} x_j(s)), \forall i \in V, \forall t \geq s+\tau$. Letting s tend to infinity from both sides of the inequality one can know $\limsup_{t\to\infty} \max_{j\in V} x_j(t) \leq p_*$.

(b) Suppose $p_* = H_*$. Then there is no doubt that $\limsup_{t\to\infty} \max_{j\in V} x_j(t) \leq p_* = H_*$

Therefore, it is almost sure that $\limsup_{t\to\infty} \max_{j\in V} x_j(t) \leq p_*$. On the other hand, $p_* > h_*$ implies that $\lim_{t\to\infty} \min_{j\in V} x_j(t) = h_*$ holds as a sure event. One can immediately draw from the structure of the algorithm that it can be only the case $\limsup_{t\to\infty} \max_{j\in V} x_j(t) = p_*$ because otherwise, there is a node i_* with $\psi_{i_*}(x_{i_*}(t)) = p_*$ for all large enough t. However, even $\limsup_{t\to\infty} \max_{j\in V} x_j(t) = p_*$ ensures that there must always be nodes whose states are arbitrarily close to p_* for an infinite amount of times, a similar contradiction argument would clarify that in that case $\lim_{t\to\infty} \min_{j\in V} x_j(t) = p_*$ holds as well. This contradicts our standing assumptions $P(p_* > h_*) > 0$. We have now proved $\mathbb{P}(p_* = h_*) = 1$.

A symmetric argument leads to $\mathbb{P}(q_* = H_*) = 1$. ■

We are now in a position to complete the proof of the main result. Define

$$w_i(t) = \varepsilon \sum_{j\in N_i(t)} a_{ij}(\psi_j(x_j(t)) - x_j(t)).$$

The algorithm (2) can be written as

$$x_i(t+1) = x_i(t) + \varepsilon \sum_{j\in N_i(t)} a_{ij}(x_j(t) - x_i(t)) + w_i(t).$$

Now from Lemma 4, one can know that there is a constant D_0 such that $|w_i(t)| \leq D_0$ is a sure event for all i and t. From the form of its definition, $w_i(t)$ is $\mathcal{F}_t$-measurable. Moreover, Lemma 5 ensures $\mathbb{P}(\lim_{t\to\infty} w_i(t) = 0) = 1$. Thus, we can obtain by applying Proposition 1 that

$$\mathbb{P}(\lim_{t\to\infty} \max_{i,j\in V} |x_i(t) - x_j(t)| = 0) = 1.$$

Finally, it is consistent with the deterministic case (Fontan et al., 2017, 2019) that $x_i(t)$ admits a finite limit within $[p_*, q_*] = \cap_{m=1}^n I_m$.

We have now completed the proof of the main theorem. ■

5. Numerical examples

We consider a network of $n = 4$ nodes indexed by $V = \{1, 2, 3, 4\}$. Let $E = \{(1, 2), (1, 4), (2, 3), (3, 4)\}$ and the associated

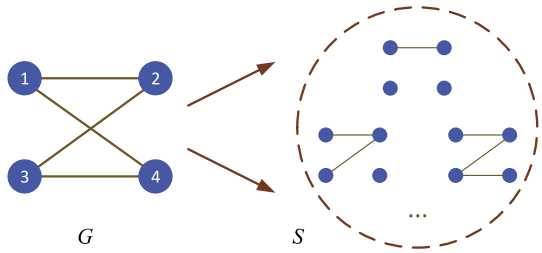
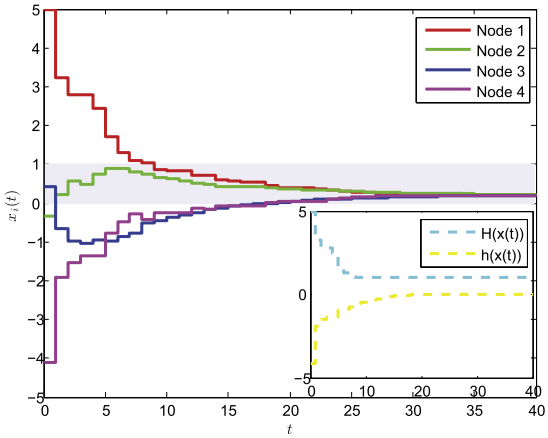
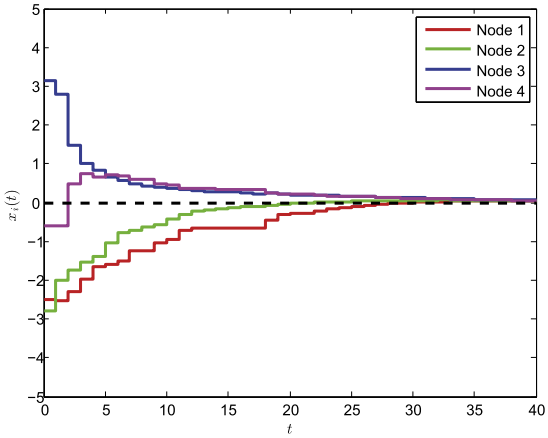


Fig. 1. The underlying graph G consisting of four nodes.



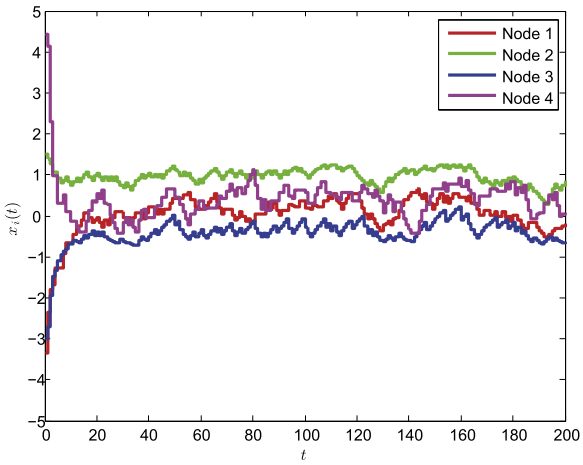
(a) All the nodes determine the consensus value



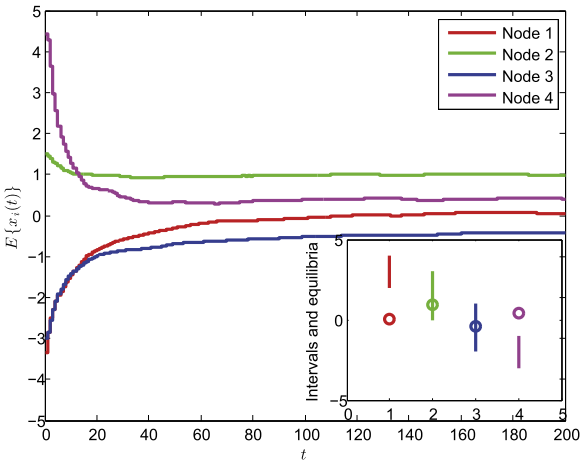
(b) One node determines the consensus value: a leader-following case

Fig. 2. Trajectories $x_i(t)$, nonempty intersection cases: (a) Example 1; (b) Example 2.

weights are $a_{12} = a_{21} = 0.5$, $a_{14} = a_{41} = 0.4$, $a_{23} = a_{32} = 0.2$, and $a_{34} = a_{43} = 1$. The underlying graph is illustrated in Fig. 1. It is apparent that G is connected. Moreover, the random variables $\mathbf{1}_{\{(i,j)\in E_{\sigma(t)}\}}$, $(i, j) \in E$, $t \in \mathbb{N}$, are i.i.d. Bernoulli ones with $\mathbb{P}((i, j) \in E_{\sigma(t)}) = 0.5$. That is, node i and node j communicate with each other with probability 0.5 at each time $t \in \mathbb{N}$ if $(i, j) \in E$. $\varepsilon = 0.3$ is also selected to ensure $2\varepsilon \sum_{j\in N_i} a_{ij} < 1$. In the following, we first consider the cases in which the intervals I_m have nonempty intersection with respectively all the nodes and only one node intending to determine the values of consensus, and then an empty intersection case.



(a) Trajectories $x_i(t)$



(b) Trajectories $\mathbb{E}\{x_i(t)\}$

Fig. 3. Trajectories $x_i(t)$ and $\mathbb{E}\{x_i(t)\}$, an empty intersection case: Example 3.

Example 1. The intervals imposed by the four nodes are respectively $I_1 = [0, 3]$, $I_2 = [-1, 2]$, $I_3 = [-2, 1]$, and $I_4 = [-3, 2]$. Therefore, one has $\cap_{m=1}^{n} I_m = [p_*, q_*] \neq \varnothing$ with $p_* = 0$ and $q_* = 1$. Fig. 2(a) is obtained by applying the interval consensus algorithm, which gives the evolution of the states of the four nodes. It can be observed that all the states of the nodes converge to a common value inside the intersection $[p_*, q_*]$ with $H(\mathbf{x}(t))$ being non-increasing and $h(\mathbf{x}(t))$ being non-decreasing.

Example 2. Here we suppose $I_1 = [0, 0]$, $I_2 = I_3 = I_4 = [-\infty, \infty]$. By using the interval consensus algorithm, we can see from Fig. 2(b) that node 1 acts as the role of a leader and all the nodes will track the constant trajectory (i.e., 0) determined by node 1. In this way, the framework of leader-following consensus problems can be contained by the interval consensus problems.

Example 3. The intervals are selected as $I_1 = [2, 4]$, $I_2 = [0, 3]$, $I_3 = [-2, 1]$, and $I_4 = [-3, -1]$ such that $\cap_{m=1}^{n} I_m$ is empty. Fig. 3(a) shows that the node states fail to converge to some fixed values when the intersection of the intervals imposed is empty. This is because each node is affected by different neighbors at each time due to sampling of the interaction graphs. However, the influences of the neighbors at different times are compensated in average, which can be seen in Fig. 3(b) that the mean of the node states is convergent, although the equilibria are different and do not fall into the interval imposed by each node.

6. Conclusions

In this paper, we have considered the interval consensus problems of discrete-time multi-agent systems under random graphs. An interval consensus algorithm has been applied to authorize each node to determine the acceptable consensus region. By using generalized robust consensus analysis, we have proved the effectiveness of the interval consensus algorithm under consideration. Numerical examples have also been provided to confirm our findings.

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